Announcement ~~10/7~~ 2/18

Assignments that are due soon

- Pre-Class Assignment Week 3-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets for Unit 1 will be closed on ~~tonight~~ 11:59PM

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

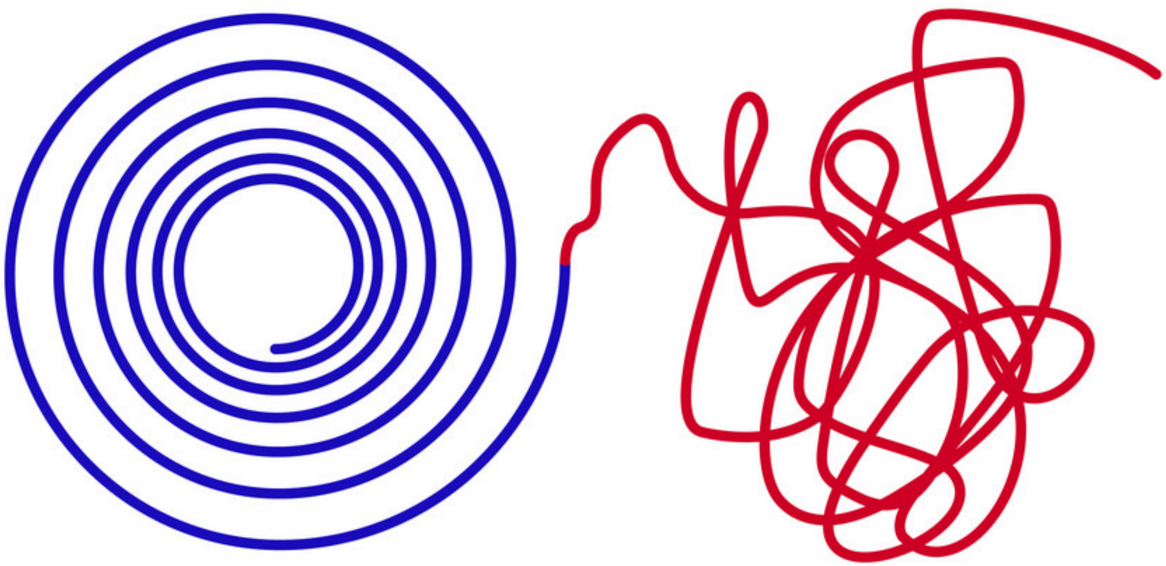
Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu ~~5~~ 4 - ~~7~~ 6 PM at Hashinger 412

CHEM 150				
Tutoring in DeMille Hall				
	Monday	Tuesday	Wednesday	Thursday
10:00 AM		Joseph		Joseph
10:30 AM		Joseph		Joseph
11:00 AM		Joseph		Joseph
11:30 AM		Joseph		Joseph
12:00 PM		Joseph		Joseph
12:30 PM		Joseph		Joseph
1:00 PM				
1:30 PM				
2:00 PM				
2:30 PM			Rachel	
3:00 PM			Rachel	
3:30 PM	Rachel		Rachel	
4:00 PM	Rachel	Malina	Rachel	
4:30 PM	Rachel	Malina	Rachel	
5:00 PM	Rachel	Malina	Rachel	
5:30 PM		Malina		

Chapter 19:
Free Energy and Thermodynamics

Order → Chaos



Dr. Wang

Learning Objectives

- Understand the concept of spontaneous change and second law of thermodynamics.
- Understand the concept of entropy and predict the sign of entropy, ΔS.
- Describe, at a basic level, the concept of a microstate.
- Calculate Gibbs Free Energy changes and predict spontaneity from ΔH and ΔS.
- Calculate standard entropy changes ΔS°.
- Calculate ΔG°_{rxn} from standard free energies of formation ΔG°_f.
- Calculate the standard change in Free Energy for a reaction using $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$.
- Determine maximum amount of energy produced by a reaction theoretically possible to do work.

Spontaneous vs Nonspontaneous

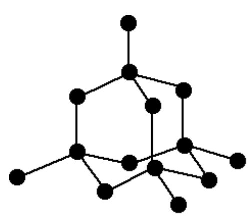
- A **spontaneous** process occurs "naturally" without ongoing external intervention.
- A **nonspontaneous** process only occurs while energy is continually supplied to the system.
- The opposite of a spontaneous process must be nonspontaneous, and vice versa.

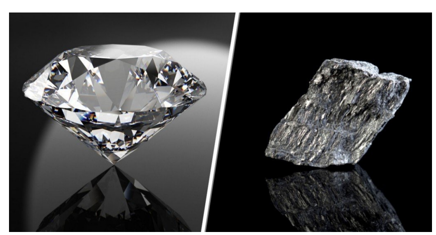
Photosynthesis:

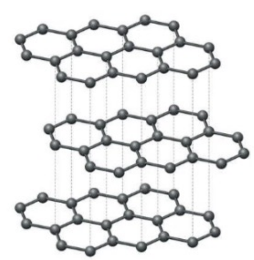
$C_6H_{12}O_6(s) + 6O_2(g) \rightleftharpoons 6CO_2(g) + 6H_2O(l)$

Respiration:

- Spontaneous ≠ Fast Speed







Diamond → Graphite

What makes a process spontaneous?

- Is spontaneous processes have to release energy and be exothermic?

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ $\Delta H^{\circ}_{rxn} = -890.4\ kJ$

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ $\Delta H^{\circ}_{rxn} = -56.2\ kJ$

- Spontaneous endothermic process:

$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq)$ $\Delta H^{\circ}_{rxn} = +25\ kJ$

$HgO(s) \rightarrow 2Hg(l) + O_2(g)$ $\Delta H^{\circ}_{rxn} = +90.7\ kJ$

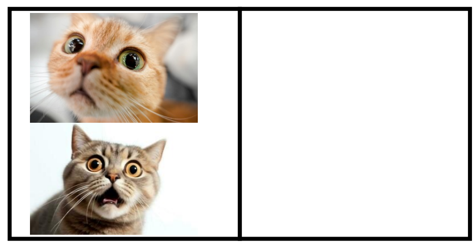
- Enthalpy is not the only factor that determines spontaneity.

Entropy

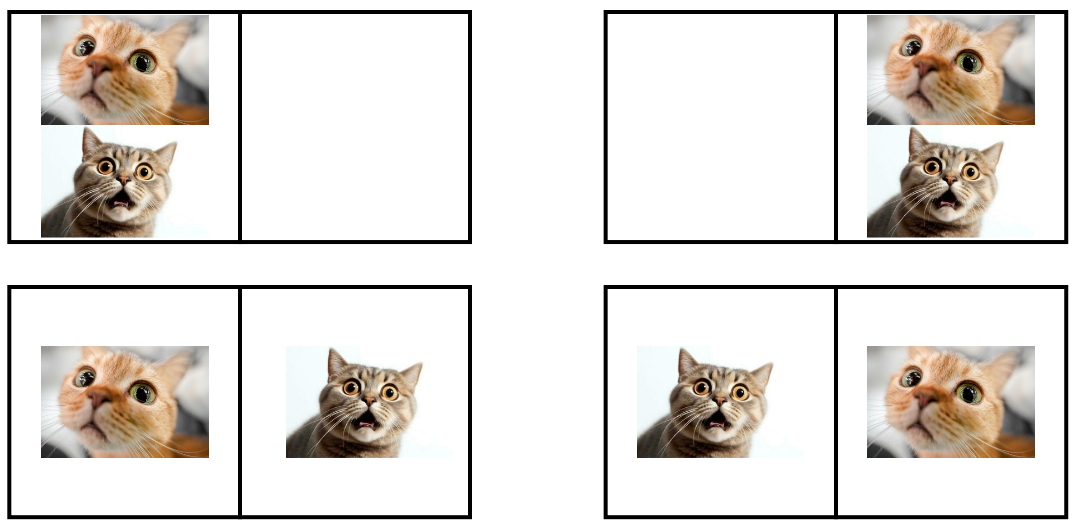
Entropy (S):

- Disorder/Randomness at molecular level.
- Microstate (W): energetically equivalent ways to arrange the particles.
- More microstates, higher entropy. $S = k\,lnW$

Only left room allowed…



Both rooms are allowed…

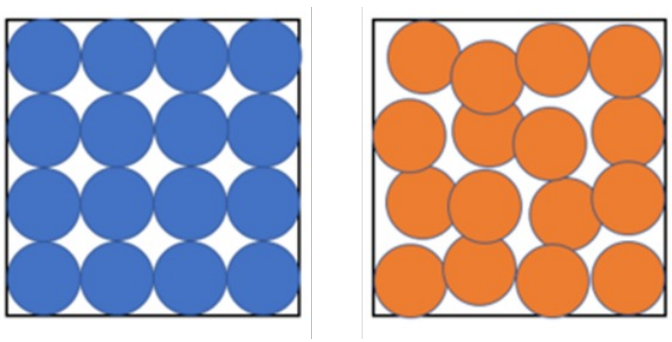


Entropy

An increase in entropy is essentially a result of increasing possible position and kinetic **energies** of molecules within the system

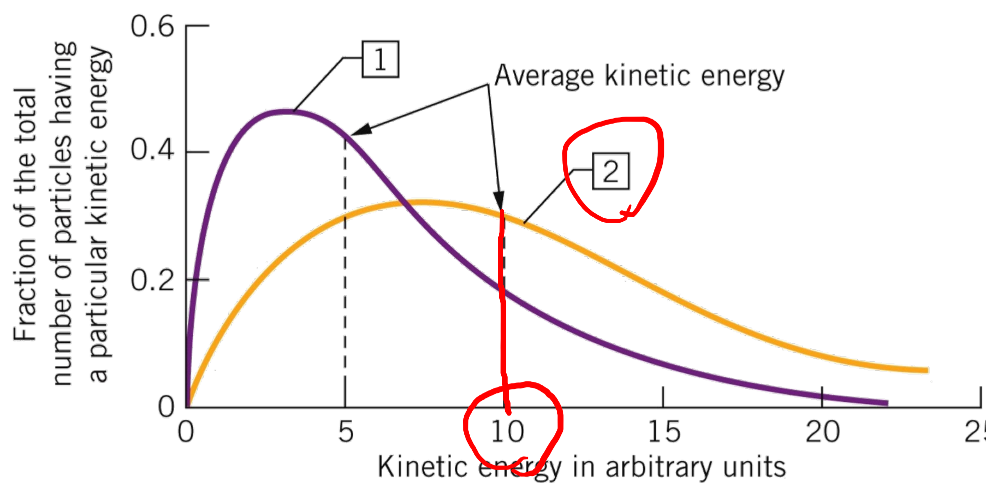
Entropy (S) increases when …

- The volume $\uparrow$
- The temperature (average kinetic energy of particles) $\uparrow$



T = 0 K
S = 0

T > 0 K
S > 0



At $\boldsymbol{T = 0\ K}$, the entropy (S) of a pure, perfect crystalline substance ($\boldsymbol{W = 1}$) is zero (commonly known as the **third law of thermodynamics**)

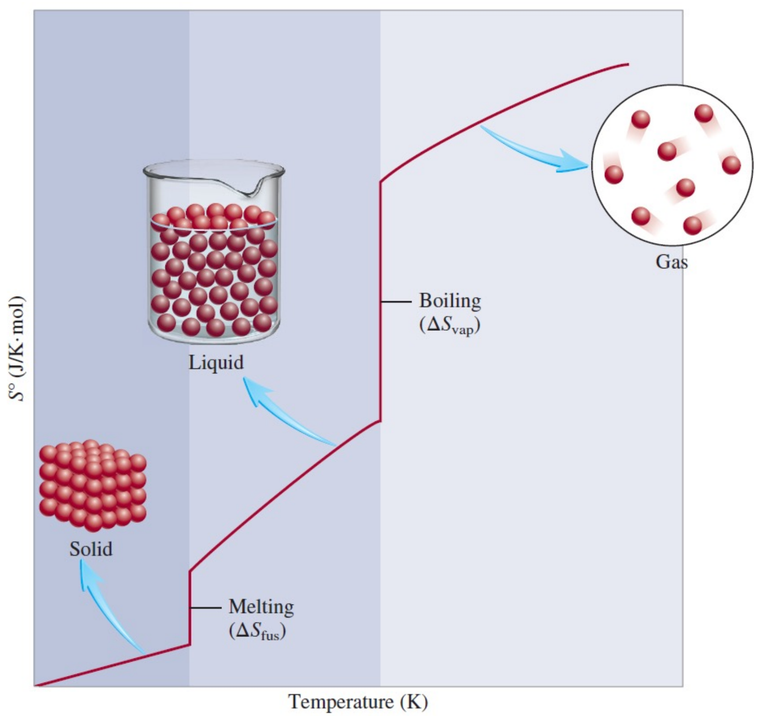
$S = k \ln 1 = 0$

Entropy

An increase in entropy is essentially a result of increasing possible **positions** and **kinetic energies** of molecules within the system

Entropy (S) increases when …

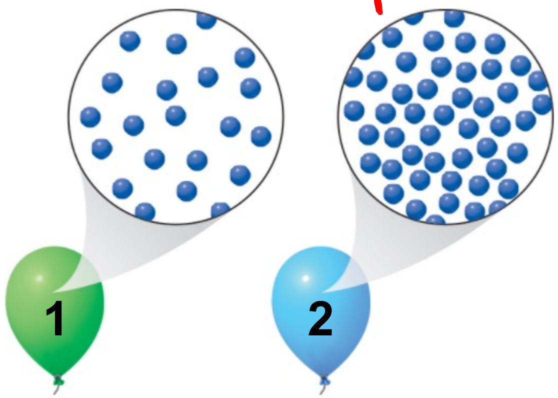
- the number of molecules ↑
- Physical state changes: S(solids) < S(liquid) << S(gas)



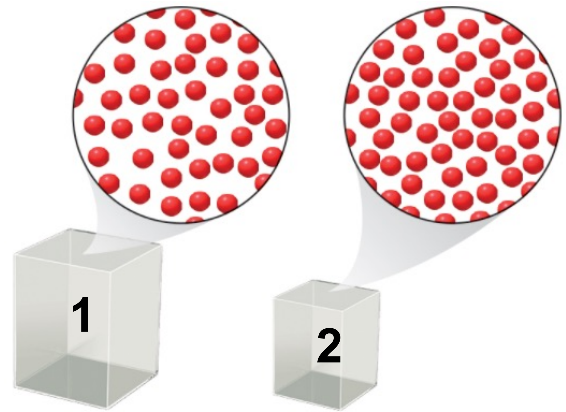
Your turn!

For the following picture, which picture depicts an entropy decrease from left to right ?

A. Same V and T,
$n_1 < n_2$ ↑ΔS (+)

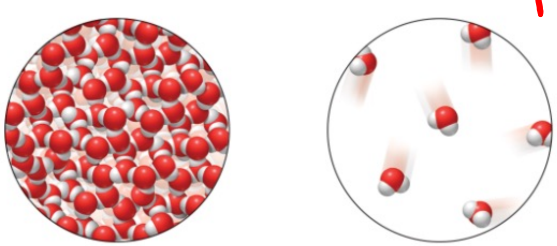


B. Same n and T,
$V_1 > V_2$ ↓



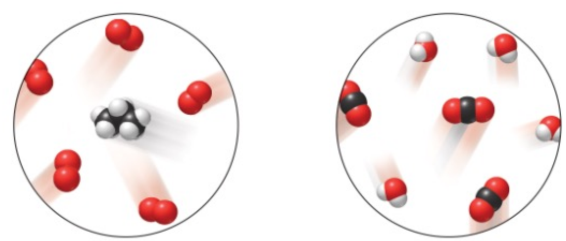
C. vaporization of water

$H_2O(l) \rightarrow H_2O(g)$ ↑



D. combustion of C_3H_8

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

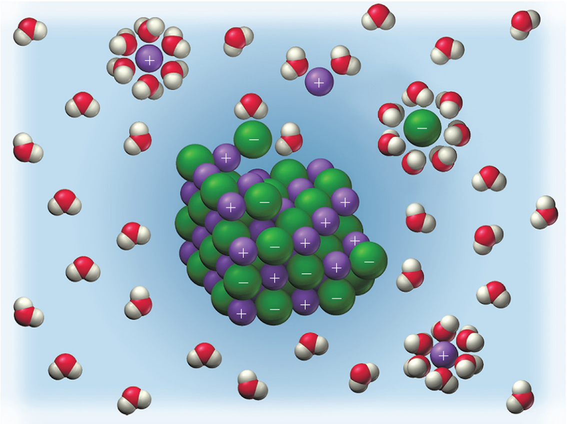


6(g)→4(g)(ℓ)

6 mol (g) → 7 mol (g)

Entropy

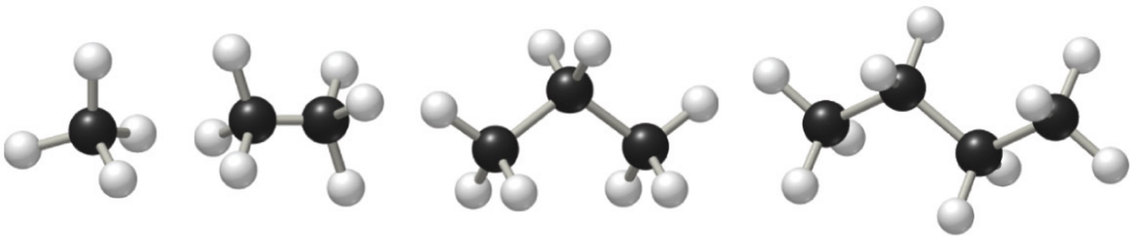
An increase in entropy is essentially a result of increasing possible **positions** and **kinetic energies** of molecules within the system



Entropy (S) increases when …

- S (pure substances) __<__ S(mixtures/solutions) (aq)
- the complexity of molecule __↑__

Formula:	CH_4	CH_3CH_3	$CH_3CH_2CH_3$	$CH_3CH_2CH_2CH_3$
$S°$, J/(mol · K):	186	230	270	310



- Entropy is a state function: $\Delta S = S_{final} - S_{initial} = S_{product} - S_{reactant}$

Your turn! $\Delta S = S_{products} - S_{reactants}$

Predict the sign of the ΔS for the following reactions. Which reaction has a negative ΔS?

A. $I_2(s) \rightarrow I_2(g)$ (+)

B. $Br_2(g) + 3\ Cl_2(g) \rightarrow 2BrCl_3(g)$ (−)

4 (g) → 2(g)

C. $Ni(s) + 2\ HCl(aq) \rightarrow H_2(g) + NiCl_2(aq)$ ↑ (+)

Ø (g) → 1 (g)

(g)

^

(aq)

^

(l)

^

(s)

Standard Entropy, S° and Entropy Change, ΔS°_{rxn}

- The absolute value of entropy, $\boldsymbol{S}$, can be determined.
- For a perfect crystalline substance at 0 K, S = $0\ \frac{J}{K}$
- For any substance at nonzero K, S is always positive
- Entropy in standard states is standard entropy ($\boldsymbol{S}^{\circ}$)
- The unit for: $\frac{J}{K \cdot mol}$

Standard Entropy Values ($S°$) for Some Substances at 25°C

Substance	$S°$ (J/K · mol)
$H_2O(l)$	69.9
$H_2O(g)$	188.7
$Br_2(l)$	152.3
$Br_2(g)$	245.3
$I_2(s)$	116.7
$I_2(g)$	260.6
C (diamond)	2.4
C (graphite)	5.69
CH_4 (methane)	186.2
C_2H_6 (ethane)	229.5
He(g)	126.1
Ne(g)	146.2

Calculate Entropy Change, ΔS°_{rxn}

Entropy is a state function:

$$\Delta S^{\circ}_{rxn} = \sum n S^{\circ}_{products} - \sum m S^{\circ}_{reactants}$$

Substance	S° (J/K · mol)
$H_2O(l)$	69.9
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$He(g)$	126.1
$Ne(g)$	146.2

Standard Entropy Values (S°) for Some Substances at 25°C

$H_2(g)$ 130.6

- $I_2(s) \rightarrow I_2(g)$

$$\Delta S^{\circ}_{rxn} = 1\,mol\,I_2(g) \times \frac{260.6\,J}{K \cdot 1\,mol\,I_2(g)} - 1\,mol\,I_2(s) \times \frac{116.7\,J}{K \cdot 1\,mol\,I_2(s)}$$

$$= +143.9\,\frac{J}{K}$$

- C (s, graphite) + 2 H_2 (g) → CH_4 (g)

$$\Delta S^{\circ}_{rxn} = 1\,mol\,CH_4(g) \cdot \frac{186.2\,J}{K \cdot 1\,mol\,CH_4(g)} - 1\,mol\,C(graphite) \cdot \frac{5.69\,J}{K \cdot mol\,C(graphite)}$$

$$- 2\,mol\,H_2(g) \cdot \frac{130.6\,J}{K \cdot 1\,mol\,H_2(g)} = -80.69\,\frac{J}{K}$$

Your turn!

Calculate **ΔS°** for the reaction below:

$Al_2O_3\ (s)\ +\ 3\ H_2\ (g)\ \rightarrow\ 2\ Al\ (s)\ +\ 3\ H_2O\ (g)$

$$\Delta S^{\circ}_{rxn} = \sum n S^{\circ}_{products} - \sum m S^{\circ}_{reactants}$$

	S° (J/K·mol)
Al (s)	28.3
Al_2O_3 (s)	51.0
H_2 (g)	130.6
H_2O (g)	188.7

179.9 J/K

Your turn!

Calculate **ΔS°** for the reaction below:

$$Al_2O_3\ (s)\ +\ 3\ H_2\ (g)\ \rightarrow\ 2\ Al\ (s)\ +\ 3\ H_2O\ (g)$$

$$\Delta S^\circ_{rxn} = \sum \textcolor{red}{n} S^\circ_{products} - \sum \textcolor{red}{m} S^\circ_{reactants}$$

$$= \left(2\,\cancel{mol\ Al} \times \frac{28.3\ J}{K \cdot 1\,\cancel{mol\ Al}} + 3\,\cancel{mol\ H_2O} \times \frac{188.7\ J}{K \cdot 1\,\cancel{mol\ H_2O}} \right)$$

$$- \left(1\,\cancel{mol\ Al_2O_3} \times \frac{51.0\ J}{K \cdot 1\,\cancel{mol\ Al_2O_3}} + 3\,\cancel{mol\ H_2} \times \frac{130.6\ J}{K \cdot 1\,\cancel{mol\ H_2}} \right)$$

$$= 179.9\ J/K$$

	S° (J/K·mol)
Al (s)	28.3
Al_2O_3 (s)	51.0
H_2 (g)	130.6
H_2O (g)	188.7

179.9 J/K

The Second Law of Thermodynamics

2nd Law of Thermodynamics:

- Spontaneous processes tend to disperse energy.

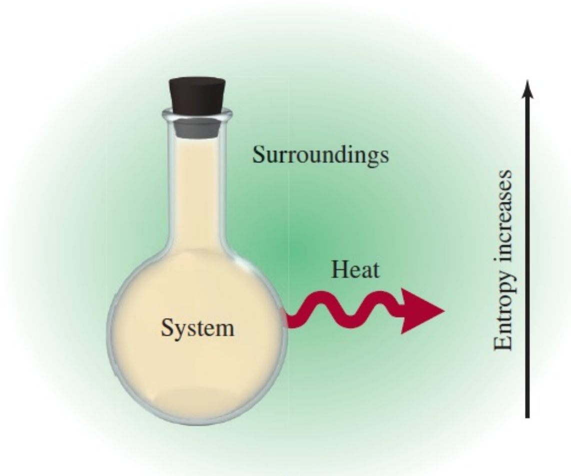
- A spontaneous event occurs when the total Entropy of universe increases.

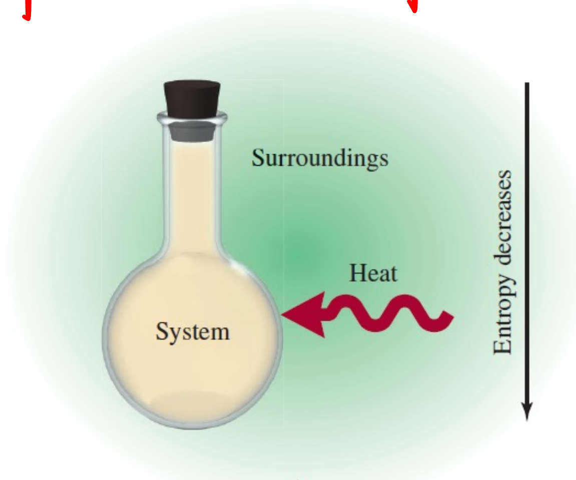
$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroudings} > 0$$

Entropy Change in the Surroundings

- At constant T, $\Delta S = \frac{q}{T}$
- Most reactions are done in atmosphere under constant pressure and temperature.

$$\Delta S_{surroundings} = \frac{q_{surroundings}}{T} = \frac{-q_{system}}{T} = \frac{-\Delta H_{system}}{T}$$





$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroudings} > 0$$

$$\Delta S_{system} - \frac{\Delta H_{system}}{T} > 0$$

$$T\Delta S_{system} - \Delta H_{system} > 0$$

Gibbs Free Energy (G)

$$\Delta G = -T\Delta S_{univ} = \Delta Hsystem - T\Delta Ssystem$$

ΔG represents the maximum energy available of a system to do work to its surroundings.

Unit of ΔG: kJ, $\frac{kJ}{mol}$

Under constant pressure and temperature:

- $\Delta G < 0$, then the reaction is spontenous
- $\Delta G = 0$, then the reaction is equibrium
- $\Delta G_{rxn} > 0$, then the reaction is nonspontenous

  (the reverse reaction should be spontenous)

- ΔG is temperature dependent.

80°F $H_2O(s) \rightarrow H_2O(l)$ ΔH = 6.01 kJ ΔS = 22.0 J/K

At 300 K, ΔG = $6.01 kJ - 300K \cdot \frac{22.0J}{K} \cdot \frac{1kJ}{10^3J} = -0.59 kJ < 0$ $\boxed{\checkmark}$

-100F°

At 200 K, ΔG = $6.01 kJ - 200K \cdot \frac{22.0J}{K} \cdot \frac{1kJ}{10^3J} = +1.61 kJ > 0$ $\boxed{\times}$

At which temperature, ΔG = 0? $6.01 kJ - T \cdot \frac{22.0J}{K} \cdot \frac{1kJ}{10^3J} = 0$

$T = 273.18 K$

$= 32.00°F$

Gibbs Free Energy (G)

$$\Delta G = -T\Delta S_{univ} = \Delta Hsystem - T\Delta Ssystem$$

ΔG represents the maximum energy available of a system to do work to its surroundings.

Under constant pressure and temperature:

- $\Delta G < 0$, then the reaction is spontaneous
- $\Delta G = 0$, then the reaction is equlibrium
- $\Delta G_{rxn} > 0$, then the reaction is nonspontaneous

  (the reverse reaction should be spontaneous)

- ΔG is temperature dependent.

$H_2O(s) \rightarrow H_2O(l)$ $\Delta H = 6.01$ kJ $\Delta S = 22.0$ J/K

At 300 K, ΔG = $6.01\,kJ - 300\,K \times \frac{22.0\,J}{K} \times \frac{1\,kJ}{10^3\,J} = -0.59\,kJ < 0$

26.85°C / 80.33°F , spontaneous

At 200 K, ΔG = $6.01\,kJ - 200\,K \times \frac{22.0\,J}{K} \times \frac{1\,kJ}{10^3\,J} = +1.61\,kJ > 0$

−73.15°C / −99.67°F , nonspontaneous

At which temperature, $\Delta G = 0$? $6.01\,kJ - T \times \frac{22.0\,J}{K} \times \frac{1\,kJ}{10^3\,J} = 0$ $T = 273.18\,K$

0.03°C / 32.05°F : melting point for water

Announcement 2/20

Exam 1 at 11 AM on 2/27:

- Due Date:

  Wed

  (2nd bonus point: midnight 2/25; 3rd +4th bonus point: 10 AM 2/27)
- 50 minutes long
- Calculators (any kind), pens/pencils/highlighters, and erasers are allowed.
- Learning objective uploaded

Assignments that are due soon

- Pre-Class Assignment Week 4-M

Office Hour

- Dr. Wang's: Fri 2-3 PM at KC 382

Your turn! $\Delta G = \Delta H_{system} - T\Delta Ssystem$

A sample of liquid ethanol must absorb 38.56 **k**J of energy to completely vaporize, and the corresponding entropy change is +109.67 J/K. Predict the boiling point of ethanol in °C. ($T_K = T_{°C} + 273.15$)

$(l) \rightarrow (g)$

→ at equilibrium $\Delta G = 0$

78.5 °C

Your turn! $\Delta G = \Delta H_{system} - T\Delta Ssystem$

A sample of liquid ethanol must absorb 38.56 **kJ** of energy to completely vaporize, and the corresponding entropy change is +109.67 J/K. Predict the boiling point of ethanol in °C. ($T_K = T_{°C} + 273.15$)

$$C_2H_5OH(l) \longrightarrow C_2H_5OH(g)$$

$$\Delta H = +38.56\,kJ \qquad \Delta S = +109.67\,\frac{J}{K}$$

$$+38.56\,kJ - T \times \frac{109.67\,J}{K} \times \frac{1\,kJ}{10^3\,J} = 0 \quad \Leftarrow 5\text{ pts}$$

$$T_F = 351.6\,K$$

$$351.6\,K = T_{°C} + 273.15$$

$$T_{°C} = 78.5\,°C \quad \Leftarrow 1\text{pt} \quad 1\text{pt unit}$$

78.5 °C

Gibbs Free Energy and Spontaneity

$$\Delta G = \Delta H_{system} (- T\Delta S system)$$

ΔH	ΔS	ΔG	Spontaneous?
− exo-	+ more chaos	(−)	✓
− exo-	−	high T, (+) low T, (−)	✗ ✓
+ endo-	+	high T, (−) low T, (+)	✓ ✗
+ endo-	− less chaos	(+)	✗

- ΔG is temperature dependent.

$H_2O(s) \rightarrow H_2O(l)$
ΔH: (+)
ΔS: (+)
ΔG: high T, ✓
low T, ✗

$H_2O(l) \rightarrow H_2O(s)$
ΔH:
ΔS:
ΔG:

Gibbs Free Energy and Spontaneity

$$\Delta G = \Delta H sy_{stem} - T\Delta S system$$

	ΔH	ΔS	ΔG	Spontaneous?
(1)	(−)	(+)	(−)	Yes
(2)	(−)	(−)	high T (+) −TΔS dominate Low T (−) ΔH dominate	No Yes
(3)	(+)	(+)	high T (−) −TΔS dominate Low T (+) ΔH dominate	Yes No
(4)	(+)	(−)	(+)	No

- ΔG is temperature dependent.

$H_2O(s) \rightarrow H_2O(l)$
ΔH: (+)
ΔS: (+)
ΔG: high temp (−)

$H_2O(l) \rightarrow H_2O(s)$
ΔH: (−)
ΔS: (−)
ΔG: low temp (−)

Your turn! $\Delta G = \Delta H_{system} - T\Delta S_{system}$

(−) (−)

For an exothermic reaction with a decrease in entropy, which temperature condition would be more likely for this reaction to occur spontaneously?

A. High temperature

B. Low temperature

C. Reaction cannot become non-spontaneous at any temperature

D. Reaction is spontaneous at any temperature

Your turn! $\Delta G = \Delta H_{system} - T\Delta Ssystem$

The following reaction is used to produce quicklime:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

Use the thermodynamic data below to 1) decide whether this reaction is spontaneous at 25 °C

Substance	ΔH_f° (kJ/mol)	S° (J/K · mol)
$CaCO_3(s)$	−1206.9	92.9
$CaO(s)$	−635.6	39.8
$CO_2(g)$	−393.5	213.6

1) nonspontaneous

Your turn! $\Delta G = \Delta H_{system} - T\Delta Ssystem$

The following reaction is used to produce quicklime:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

Use the thermodynamic data below to 2) calculate the temperature, °C, at which this reaction changes from nonspontaneous to spontaneous. ($T_K = T_{°C} + 273.15$)

Substance	ΔH_f° (kJ/mol)	S° (J/K · mol)
$CaCO_3(s)$	−1206.9	92.9
$CaO(s)$	−635.6	39.8
$CO_2(g)$	−393.5	213.6

2) 835 ºC

Your turn! $\Delta G = \Delta H_{system} - T\Delta S system$

The following reaction is used to produce quicklime:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

Use the thermodynamic data below to 1) decide whether this reaction is spontaneous at 25 °C, and 2) calculate the temperature, °C, at which this reaction changes from nonspontaneous to spontaneous. ($T_K = T_{°C} + 273.15$)

Substance	ΔH_f° (kJ/mol)	S° (J/K · mol)
$CaCO_3(s)$	−1206.9	92.9
$CaO(s)$	−635.6	39.8
$CO_2(g)$	−393.5	213.6

1) $\Delta H_{rxn} = 1\text{ mol CaO} \times \frac{-635.6\text{ kJ}}{1\text{ mol CaO}} + 1\text{ mol }CO_2 \times \frac{-393.5\text{ kJ}}{1\text{ mol }CO_2} - 1\text{ mol }CaCO_3 \times \frac{-1206.9\text{ kJ}}{1\text{ mol }CaCO_3}$

$= +177.8\text{ kJ} \Leftarrow$

$\Delta S_{rxn} = 1\text{ mol CaO} \times \frac{39.8\text{ J}}{1\text{ mol CaO}\cdot K} + 1\text{ mol }CO_2 \times \frac{213.6\text{ J}}{1\text{ mol }CO_2\cdot K} - 1\text{ mol }CaCO_3 \times \frac{92.9\text{ J}}{1\text{ mol }CaCO_3\cdot K}$

$= +160.5\text{ J/K} \Leftarrow$

$\Delta G_{rxn} = +177.8\text{ kJ} - (25+273.15)\text{ K} \times \frac{160.5\text{ J}}{K} \times \frac{1\text{ kJ}}{10^3\text{ J}} = +130\text{ kJ} > 0$

non spontaneous

2) $177.8\text{ kJ} - T \times 160.5\text{ J} \times \frac{1\text{ kJ}}{10^3\text{ J}} = 0$

$T = 1108\text{ K}$

$T_{°C} = 1108\text{ K} - 273.15 = 835\,°C$

1) nonspontaneous; 2) 835 oC

Calculate ΔG°_{rxn} using Formation Reactions

- ΔG is a state function.
- ΔG°_{f} is called standard free-energy of formation.
- Unit for ΔG°_{f} : _________
- For any any pure element in its standard state: ΔG°_{f} = ___ kJ/mol

$$\Delta G^{\circ}_{rxn} = \sum \color{red}{n} \color{black}{\Delta G^{\circ}_{f,products}} - \sum \color{red}{m} \color{black}{\Delta G^{\circ}_{f,reactants}}$$

Al_2O_3 (s) + 3 H_2(g) → 2 Al(s) + 3 H_2O (g)

$\Delta G^{\circ}_{rxn} =$

Substance	ΔH_f° (kJ/mol)
Al_2O_3 (s)	-1669.8
H_2O (g)	-241.8

Your turn!

$$\Delta G^{\circ}_{rxn} = \sum n \Delta G^{\circ}_{f,products} - \sum m \Delta G^{\circ}_{f,reactants}$$

Calculate the maximum work, in kJ, that the combustion of 125 g ethanol, C_2H_5OH (l) (mm = 46.068 g/mol), could do.

$$C_2H_5OH\ (l) + 3\ O_2(g) \rightarrow 2\ CO_2\ (g) + 3\ H_2O\ (l)$$

	ΔG_f° (kJ/mol)
C_2H_5OH (l)	-174.8
H_2O (l)	-237.2
CO_2 (g)	-394.4

-3597 kJ

Summary: Three Laws of Thermodynamics

- **The first law of thermodynamics:**

  energy cannot be created or destroyed but can only be converted from one form to another.

- **The second law of thermodynamics:**

  Spontaneous processes tend to disperse the energy.

- **The third law of thermodynamics:**

  At 0 K, a pure, perfect crystalline substance has an entropy of zero